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Sayan Biswas

Research interests

Trustworthy and Collaborative Machine Learning (focusing on Privacy, Adversarial Robustness, and Fairness).

Positions

2023 - present **Postdoctoral Researcher** at EPFL (Scalable Computing Systems Lab) – Lausanne, Switzerland. Supervisor: Prof. Anne-Marie Kermarrec.

Education

2020 - 2023 **PhD in Computer Science** from INRIA, École Polytechnique, and Institut Polytechnique de Paris – Palaiseau, France.
Supervisor: Prof. Catuscia Palamidessi. **Best PhD Thesis in Computer Science (accessit award)** from IP Paris.

2016 - 2020 **Master of Mathematics (M.Math.)** with **First-Class Honours** from University of Bath – Bath, England.

Grants and Funding

2025 **Grant of CHF 65,452.00 from the Federal Office for Defence Procurement of Switzerland and *armasuisse*** for the project "Privacy-preserving and distributed processing of public data in hybrid trust networks".

Awards and Honours

2026 **Invited participant** at **Dagstuhl Seminar 26401 on "Divided We Stand: Security and Privacy of Collaborative Learning"** held at Schloss Dagstuhl - Leibniz Center for Informatics in Wadern, Germany.

2025 **Young Scholars' Development Program Fellowship** from ACM CCS'25.

2024 **Accessit Award for the Best PhD Thesis in Computer Science** from Institut Polytechnique de Paris, France.

Publications

Conference Papers

[C14] **Your Neighbors Know: Leveraging Local Neighborhoods for Backdoor Detection in Decentralized Learning.**
Sayan Biswas, Antoine Boutet, Davide Frey, Romaric Gaudel, Rachid Guerraoui, Maxime Jacovella, Anne-Marie Kermarrec, Dimitri Lerévérend, François Taïani, and Martijn de Vos.
Advances in Neural Information Processing Systems (NeurIPS), 2026.

[C13] **Federated and Differentially Private Estimation of KL Divergence.**
Sayan Biswas, Graham Cormode, Carsten Maple, and Mary Scott.
IEEE International Conference on Big Data (IEEE BigData), 2026.

[C12] **Communication-Efficient Secure Aggregation in Decentralized Learning.**
Sayan Biswas, Anne-Marie Kermarrec, Rafael Pires, Rishi Sharma, and Milos Vujasinovic.
IEEE International Symposium on Reliable Distributed Systems (SRDS), 2026.

[C11] **Robust ML Auditing using Prior Knowledge.**
Jade Garcia Bourrée, Augustin Godinot, Martijn de Vos, Milos Vujasinovic, **Sayan Biswas**, Gilles Tredan, Erwan Le Merrer, and Anne-Marie Kermarrec.
International Conference on Machine Learning (ICML), 2025. **Spotlight** (top 2.6% of submissions).
Also presented at the Technical AI Governance (TAIG) Workshop at ICML 2025.

[C10] **Low-Cost Privacy-Aware Decentralized Learning.**
Sayan Biswas, Davide Frey, Romaric Gaudel, Anne-Marie Kermarrec, Dimitri Lerévérend, Rafael Pires, Rishi Sharma, and François Taïani.
Proceedings on Privacy Enhancing Technologies (PoPETs), 2025, Issue 3.

[C9] **Noiseless Privacy-Preserving Decentralized Learning.**
Sayan Biswas, Mathieu Even, Anne-Marie Kermarrec, Laurent Massoulié, Rafael Pires, Rishi Sharma, and Martijn de Vos.
Proceedings on Privacy Enhancing Technologies (PoPETs), 2025, Issue 1.

[C8] **Boosting Asynchronous Decentralized Learning with Model Fragmentation.**
Sayan Biswas, Anne-Marie Kermarrec, Alexis Marouani, Rafael Pires, Rishi Sharma, and Martijn de Vos.
The ACM Web Conference (WWW), 2025. **Oral presentation** (top 7% of papers).

[C7] **Fair Decentralized Learning.**
Sayan Biswas, Anne-Marie Kermarrec, Rishi Sharma, Thibaud Trinca, and Martijn de Vos.
IEEE Conference on Secure and Trustworthy Machine Learning (SaTML), 2025.

[C6] **PRIVIC: A Privacy-Preserving Method for Incremental Collection of Location Data.**
Sayan Biswas and Catuscia Palamidessi.
Proceedings on Privacy Enhancing Technologies (PoPETs), 2024, Issue 1.
Also presented at the Theory and Practice of Differential Privacy Workshop (TPDP) 2023.

[C5] **Tight Differential Privacy Guarantees for the Shuffle Model with *k*-Randomized Response.**
Sayan Biswas, Kangsoo Jung, and Catuscia Palamidessi.
International Symposium on Foundations and Practice of Security (FPS), 2023.

[C4] **Group Privacy for Personalized Federated Learning.**
Filippo Galli, **Sayan Biswas**, Kangsoo Jung, Tommaso Cucinotta, and Catuscia Palamidessi.
International Conference on Information Systems Security and Privacy (ICISSP), 2023.
Preliminary version at the FL-NeurIPS 2022 Workshop. **Oral presentation** (top 10% of papers).

[C3] **Impact of Sampling on Locally Differentially Private Data Collection.**
Sayan Biswas, Graham Cormode, and Carsten Maple.
Competitive Advantage in the Digital Economy (CADE), 2022. **Best Paper Award**.

[C2] **Tight Differential Privacy Blanket for Shuffle Model**
Sayan Biswas, Kangsoo Jung, and Catuscia Palamidessi.
Competitive Advantage in the Digital Economy (CADE), 2022.

[C1] **An Incentive Mechanism for Trading Personal Data in Data Markets.**
Sayan Biswas, Kangsoo Jung, and Catuscia Palamidessi.
International Colloquium on Theoretical Aspects of Computing (ICTAC), 2021.

Journal Articles

[J3] **Practical and Private Hybrid ML Inference with Fully Homomorphic Encryption.**
Sayan Biswas, Philippe Chartier, Akash Dhasade, Tom Jurien de la Gravière, David Kerriou, Anne-Marie Kermarrec, Mohammed Lemou, Franklin Tranie, Martijn de Vos, and Milos Vujasinovic.
Transactions on Machine Learning Research (TMLR), 2026.

[J2] **A Privacy-Preserving Querying Mechanism with High Utility for Electric Vehicles.**
Ugur Ilker Atmaca, **Sayan Biswas**, Carsten Maple, and Catuscia Palamidessi
IEEE Open Journal of Vehicular Technology, vol. 5, 2024.

[J1] **Advancing Personalized Federated Learning: Group Privacy, Fairness, and Beyond.**
Filippo Galli, Kangsoo Jung, **Sayan Biswas**, Catuscia Palamidessi, and Tommaso Cucinotta
SN Computer Science, vol. 4, no. 6, 2023.

Book Chapter

[B1] **Establishing the Price of Privacy in Federated Data Trading.**
Kangsoo Jung, **Sayan Biswas**, and Catuscia Palamidessi
In *Protocols, Strands, and Logic*, LNCS 13066, pp. 232–250, Springer, 2021.

Workshop Papers

[W3] **Mitigating Membership Inference Vulnerability in Iterative Federated Clustering Algorithm.**
Kangsoo Jung, **Sayan Biswas**, and Catuscia Palamidessi
Workshop on Recent Advances in Resilient and Trustworthy Machine Learning-Driven Systems (ARTMAN), co-located with ACM CCS, 2025.

[W2] **Bayes' Capacity as a Measure for Reconstruction Attacks in Federated Learning.**
Sayan Biswas, Mark Dras, Pedro Faustini, Natasha Fernandes, Annabelle McIver, Catuscia Palamidessi, and Parastoo Sadeghi
Workshop on Security, Privacy, and Information Theory (Protect-IT), co-located with IEEE CSF, 2024. Non-archival.

[W1] **On the Adaptive Sensitivity of Differentially Private Machine Learning.**
Filippo Galli, **Sayan Biswas**, Kangsoo Jung, Tommaso Cucinotta, and Catuscia Palamidessi
AAAI Workshop on Privacy-Preserving Artificial Intelligence (PPAI), 2023. Non-archival.

Selected Talks

Sep 2026 **Building Neighbourhoods That Keep Secrets and Catch Trojans: Privacy and Robustness Native to Decentralized ML**
Dagstuhl Seminar 26401 "Divided We Stand: Security and Privacy of Collaborative Learning", Schloss Dagstuhl, Germany.

Mar 2026 **Decentralised ML: Trust No One, Train Together**
Foundations of AI Security and Privacy Workshop, Swiss AI Days 2026, Fribourg, Switzerland.

Oct 2022 **Incremental Collection of Location Data: Differential Privacy and Beyond...**
7th Franco-Japanese Cybersecurity Workshop, Tokyo, Japan.

Teaching

2022 **Instructor**, Inria–DFKI European Summer School on AI (IDESSAI 2022), Saarbrücken, Germany. Designed and taught the hands-on course *Differential Privacy and Federated Learning: Theory and Implementation*; **voted the most-liked course** of the school.

2021–2023 **Teaching Assistant**, École Polytechnique, Palaiseau, France: CSE 102 *Advanced Programming with Python*.

2018–2020 **Teaching Assistant**, University of Bath, UK: XX10190 *Programming and Discrete Mathematics with MATLAB* (2018–19); MA10209 *Algebra 1A* and MA10212 *Probability and Statistics 1B* (2019–20).

Academic Service

Conference PC: ACM WWW 2027, NeurIPS 2026, IEEE SaTML 2026, ACM CCS 2026, IEEE ICDCS 2026, AISTATS 2025, IEEE SaTML 2025.

Workshop PC: InfPriv Workshop at NeurIPS 2026; Workshop on Secure and Trustworthy Machine Unlearning Systems at ESORICS 2025; AAAI Workshop on Privacy-Preserving AI (PPAI) 2023.

Ad hoc reviewer for journals: The VLDB Journal, ACM Transactions on Privacy and Security, IEEE Journal on Selected Areas in Information Theory, IEEE Transactions on Dependable and Secure Computing.

Research Visits and Internships

Sep–Oct 2023 **Visiting Scholar**, Macquarie University, Sydney, Australia. Hosts: Prof. Annabelle McIver and Dr. Natasha Fernandes.

Jan–Mar 2022 **Visiting Scholar**, University of Warwick, Coventry, UK. Hosts: Prof. Carsten Maple and Prof. Graham Cormode.

Jun–Aug 2020 **Research Intern**, University of Warwick, Coventry, UK. Supervisors: Prof. Graham Cormode and Prof. Carsten Maple.

Jun–Sep 2019 **Research Intern**, Inria, Palaiseau, France. Supervisor: Prof. Catuscia Palamidessi.

Jun–Aug 2018 **Research Intern**, Institute for Mathematical Innovation, University of Bath, UK. Supervisor: Prof. Christopher Jennison.

Outreach and Leadership

2018–2020 **Lead student editor**, Department of Mathematics newsletter, University of Bath.

2017–2020 **Volunteer**, United Kingdom Mathematics Trust, and **mentor**, British Mathematical Olympiad.

2019 **Invigilator and examiner**, International Mathematical Olympiad (IMO 2019), Bath, UK.

2017–2019 **Project coordinator and head**, Mathscon, the UK's largest student-led mathematics conference.

2017 **Foreign exchange teacher**, Humen Foreign Language School (HFLS) International Summer Camp, China: taught mathematics and English.

Earlier Distinctions

2018 Qualified ICPC North-Western European Regional Contest (NWERC 2018) for ICPC European Finals 2018, representing the University of Bath; **first solver of Problem 5 in the UK and Ireland** qualifying round.

2013–2016 Pre-university olympiads (India):
Honourable Mention in the final round of the *Indian National Philosophy Olympiad 2016* **(top 10 in India)**.
Honourable Mention at the *International Linguistics Olympiad Team Selection Training Camp 2015*, the invitational round for Indian team selection after qualifying the national stage **(top 15 in India)**.
Qualified *Zonal Informatics Olympiad 2016* to represent West Bengal, India, at the *Indian National Informatics Olympiad* **(top 6 in West Bengal)**.
Qualified *Regional Mathematical Olympiad 2013 and 2014* to represent West Bengal, India, at the *Indian National Mathematical Olympiad* **(top 28 in West Bengal)**.